

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S3 (R,S) / S1 (PT) (S, FE) Examination December 2023 (2019 scheme)

Course Code: MAT201**Course Name: Partial Differential equations and Complex analysis**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 Derive partial differential equation from $z = f(x + 2t) + g(x - 2t)$ (3)
- 2 Solve $xp + yq = z$. (3)
- 3 Find the steady state temperature distribution in a rod of length 30 cm if the ends are kept at $10^0 C$ and $100^0 C$. (3)
- 4 A tightly stretched string of length 50 cm has its ends fastened at $x = 0$ and $x = 50$. The midpoint of the string is then taken to height h and then released from rest in that position. Find the initial position of the string. (3)
- 5 Check whether the function $f(z) = \begin{cases} \frac{Re(z)}{1-|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ is continuous at $z = 0$. (3)
- 6 Is $f(z) = \bar{z}$ analytic. Justify your answer. (3)
- 7 Evaluate $\int_0^{1+i} (x - iy) dz$ along the straight line path from 0 to $1+i$ (3)
- 8 Evaluate $\int_C \frac{dz}{(z^2+1)(z-1)}$ when C is $|z| = 0.5$ (3)
- 9 Find the residue of $z \cos\left(\frac{1}{z}\right)$ at $z = 0$ (3)
- 10 State Cauchy's integral formula and Cauchy's residue theorem (3)

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Form the partial differential equation by eliminating the arbitrary function f from the relation $xyz = f(x + y + z)$. (7)

- (b) Solve $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u$ with the initial condition $u(x, 0) = e^{2x}$ by applying method of separation of variables. (7)
- 12 (a) Solve $\frac{\partial^2 u}{\partial x \partial y} = \cos(x) \cos(y)$ given that $\frac{\partial u}{\partial y} = e^{-y}$ when $x = 0$ and $u = 0$ when $y = 0$. (7)
- (b) Solve $(y^2 + z^2)p - xyq + xz = 0$ (7)

Module 2

- 13 (a) Derive one-dimensional wave equation. (7)
- (b) Solve the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(0, t) = 0^\circ\text{C}$, $u(1, t) = 0^\circ\text{C}$ and the temperature initially is $u(x, 0) = x(1 - x)$. (7)
- 14 (a) A homogeneous rod of conducting material of length l has its ends kept at zero temperature and the temperature initially is $u(x, 0) = f(x)$. Find the temperature $u(x, t)$ at any time. (7)
- (b) Solve the one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to $u(0, t) = 0$, $u(1, t) = 0$, $u(x, 0) = 3\sin(2\pi x)$, $\frac{\partial u}{\partial t}(x, 0) = 0$. (7)

Module 3

- 15 (a) Find all points where the Cauchy-Riemann equations are satisfied for the function $f(z) = xy$ and where it is analytic. Give reasoning for your answers. (7)
- (b) Show that $u = 2 + 3x - y + x^2 - y^2 - 4xy$ is harmonic. Also find its conjugate harmonic function v . (7)
- 16 (a) Show that the function $f(z) = \frac{1}{z}$ is analytic except at $z = 0$. (7)
- (b) Find the image of the triangular region bounded by $x = 1$, $y = 1$ and $x + y = 1$ under the transformation $w = z^2$. (7)

Module 4

- 17 (a) Evaluate $\int_C \frac{1}{(z+4i)(z+i)} dz$, where C is $|z| = 2$ using Cauchy's integral formula. (7)
- (b) Find the Maclaurin series expansion of $e^z \sin z$. (7)
- 18 (a) Evaluate $\int_C \frac{z^2 + z + 1}{(z-1)^2} dz$, where C is the circle $|z| = 3$ using Cauchy's integral formula. (7)
- (b) Find the Taylor series expansion of $f(z) = \frac{1}{z^2}$ about $z = 2$. (7)

Module 5

- 19 (a) Obtain the Laurent series expansion of $\frac{1}{(z-2)(z-3)}$ in the region (7)
 (i) $|z| < 1$ (ii) $2 < |z| < 3$ (iii) $|z| > 3$
- (b) Evaluate $\int_0^\infty \frac{1}{1+x^2} dx$. (7)
- 20 (a) Find the poles and residues of the function $\frac{1}{z^4-1}$ (7)
 (b) Evaluate $\int_0^{2\pi} \frac{1}{3+2\sin\theta} d\theta$ (7)